DE 424: Practical 10 Report

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Declare that this work presented is your own work. Mention any sources that contributed to work or report I declare that this report is my own work. This report was written using LaTeX and thus Gemini was used to assist in certain formatting. This includes help to format code listings, mathematical equations and image display. The mathematical derivations and interpretation of results are my own work.

ECSA graduate attributes

Discuss how the GAs were assessed in the module + which parts of the report this is documented. State HOW and WHERE they were satisfied This section may (and should) be brief, but should discuss all aspects of the GAs being assessed. For example, GA1 partially assesses your ability to research literature as part of problem solving, which means that your report must mention and cite (at least) a few sources - can include prescribed resources. Please consult the module framework for a more detailed description of these two GAs as well as how they are assessed in this module.

GA 1: Problem solving

GA 2: Application of scientific and engineering knowledge

1 Problem Description

This project A monster - called the wumpus - moves throughout a grid world. It moves randomly either one cell up, down, left or right, from one time step to the next. Only remaining in its position if it attempts to move outside the boundary of the grid world. To our dismay, the exact wumpus location is unknown (latent), however known detection data is generated by each cell for each time step. At each time step and for every cell in the grid world, the wumpus is correctly detected with a probability ρ_w, and is falsely detected with probability ρ_c, which is known as a clutter measurement. Given detection data across multiple timesteps, these observations can be used as evidence with the aim to design, implement and evaluate a PGM-based solution to infer the wumpus location across all time steps for different datasets.

2 Solution Design

Model structure and show all factors. State and motivate all design choices

2.1 Dataset 1

A 5×5 grid, 10 time steps, known parameters $\rho_w = 0.95$ and $\rho_c = 0.05$

3 Code Implementation

Discussion + code snippets

4

Description of how you processed the output of solution, the presentation of results and analysis thereof. Report both qualitative and quantitative results.